

WASP-36b

R-2261

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_131216 · created 2026-08-16T01:19:28+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456642.91954 +/- 0.00086 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.147 +/- 0.011
Transit depth (Rp/Rs)^2	2.17 +/- 0.32 [%]
Orbital Inclination (inc)	85.92 +/- 1.23
Airmass coefficient 1 (a1)	1.008 +/- 0.014
Airmass coefficient 2 (a2)	-0.005 +/- 0.027
Scatter in the residuals of the lightcurve fit is	1.42 %
Best Comparison Star	#2 - [93, 32]
Optimal Method	PSF photometry
Transit Duration (day)	0.0896 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.147 ± 0.011 vs 0.1368 (deviation 0.79σ)
Duration fit vs published	0.0896 d vs 0.0758 d (deviation 2.8σ)
Mid-time O-C	0.6 min
Depth SNR	11.4
Residual scatter	1.42 %

Light curve

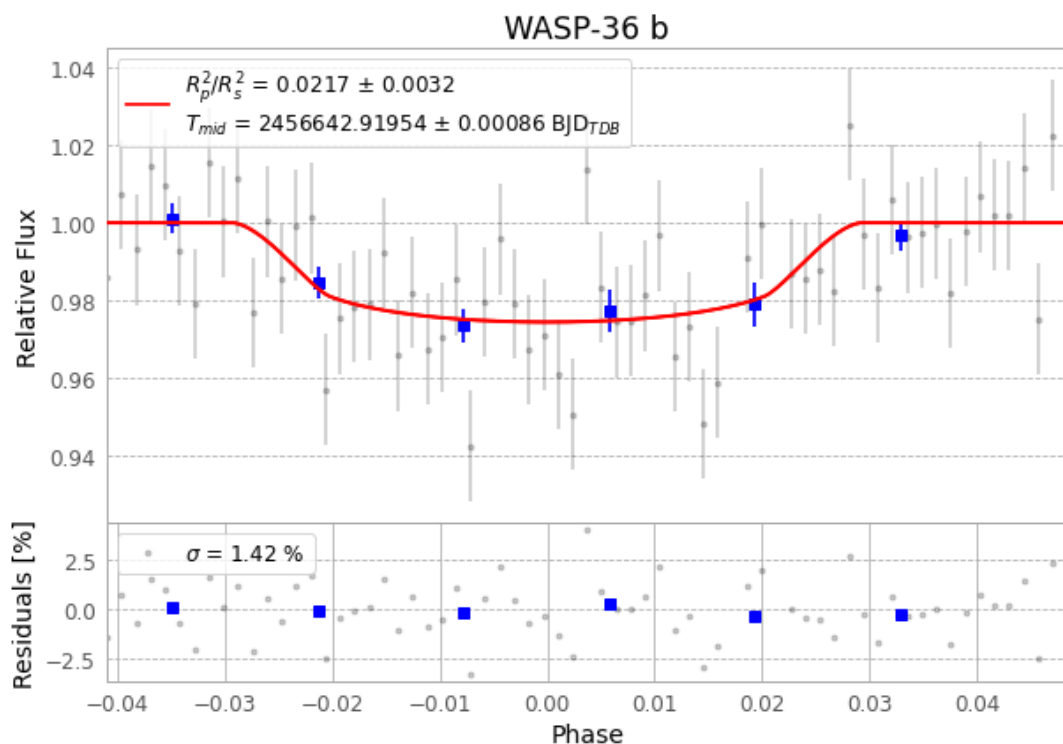


Figure 1 — FinalLightCurve_WASP-36 b_2013-12-16.png (SHA-256 e3db2f85078e025a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [228, 245], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ee8c9a1cc630921e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

67 FITS frames (44 MB), WASP-36131216082712.FITS → WASP-36131216114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-131216.log (8bb0a8134194...), FinalLightCurve_WASP-36 b_2013-12-16.png (e3db2f85078e...), FinalLightCurve_WASP-36 b_2013-12-16.csv (6801614eed0...)

Compute resources

Wall time 138.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 01:19Z.